TRAMA3 T51 — Interfaz simetrica A/B
Revision 102 conservada. T07/T08/T52 pendientes.
Esquema documental por componentes y redes globales.
Coordenadas y cableado fisico: hardware/diseno-placas/soldadura.
Puentes/switches: conexiones internas definidas en la guia y CSV.

Alimentacion y proteccion



File: 01_POWER.kicad_sch

MCP23017 e I2C



File: 02_CORE.kicad_sch

Transmisores luz y sonido



File: 03_TX.kicad_sch

Servo y corte de mantenimiento



File: 04_SERVO.kicad_sch

Receptores y puntos de prueba



File: 05_RX.kicad_sch

Selectores de fallo y enlaces GPIO



File: 06_ENABLE.kicad_sch

Reloj aislado TX y RX



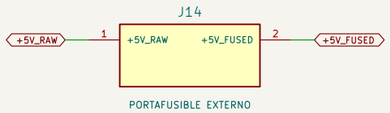
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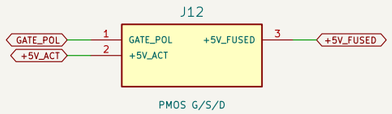
Sheet: /
File: TRAMA3_T51.kicad_sch

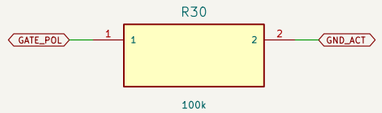
Title: TRAMA3 T51 — Indice de circuitos A/B

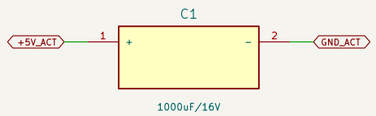
Size: A4	Date:	**Rev: P1 — PROVISIONAL**
KiCad E.D.A. 10.0.8		Id: 1/8

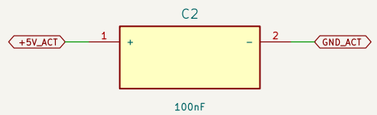


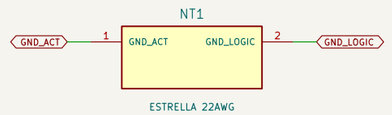


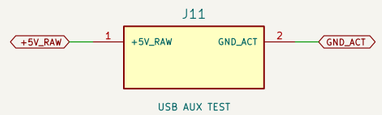


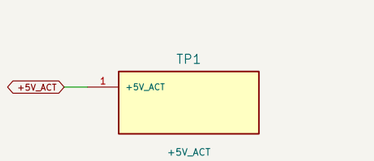


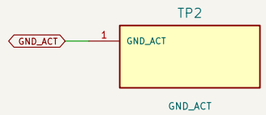


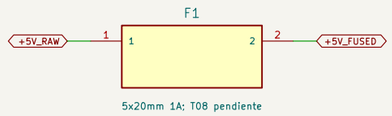


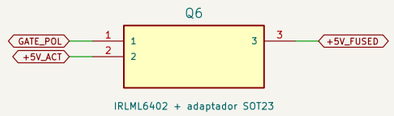


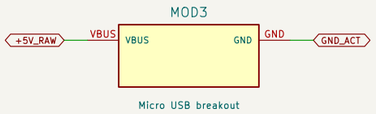












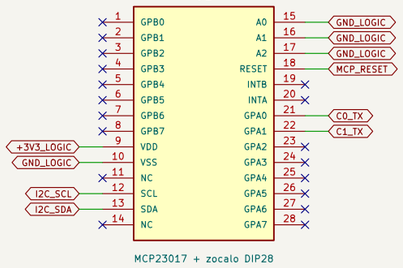
J1 o J11, nunca simultaneos. NT1: puente local 1–2. Q6: D entrada / S salida / G a R30. F1 pendiente T08.

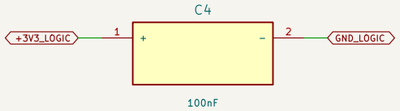
Etiquetas globales del mismo nombre son la misma red. No autoriza soldadura.

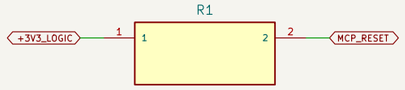
Sheet: /Alimentacion y proteccion/
File: 01_POWER.kicad_sch

Title: Alimentacion y proteccion — Nodo A/B

Size: A3	Date:	**Rev: T51–P1 PROVISIONAL**
KiCad E.D.A. 10.0.8		Id: 2/8



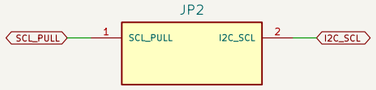


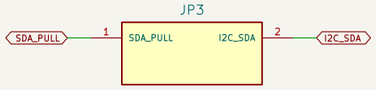


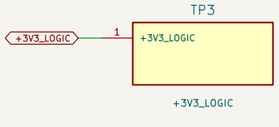


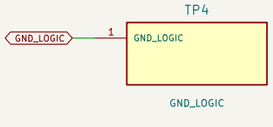












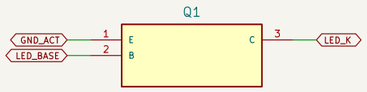
U1 muesca hacia fila A. R2/R3 DNP y JP2/JP3 abiertos. P9_3 alimenta logica 3.3V; NO 5V desde BBB.

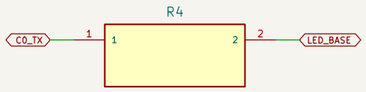
Etiquetas globales del mismo nombre son la misma red. No autoriza soldadura.

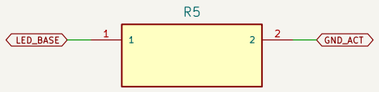
Sheet: /MCP23017 e I2C/
File: 02_CORE.kicad_sch

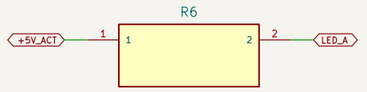
Title: MCP23017 e I2C — Nodo A/B

Size: A3	Date:	**Rev: T51-P1 PROVISIONAL**
KiCad E.D.A. 10.0.8		Id: 3/8

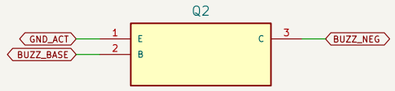


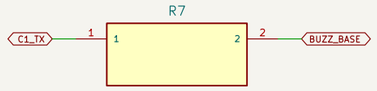














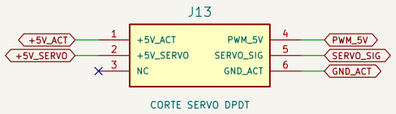
Q1/Q2 pads E−B−C; verificar lote. LED A/K y D1 banda a +5V. Buzzer: 5V y <=20mA provisional.

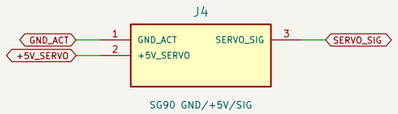
Etiquetas globales del mismo nombre son la misma red. No autoriza soldadura.

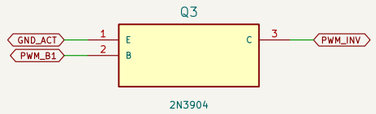
Sheet: /Transmisores luz y sonido/
File: 03_TX.kicad_sch

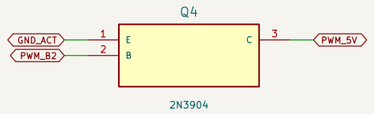
Title: Transmisores luz y sonido — Nodo A/B

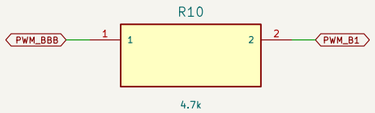
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KiCad E.D.A. 10.0.8		Id: 4/8

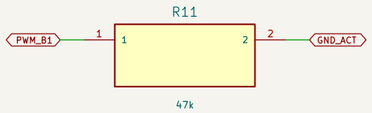


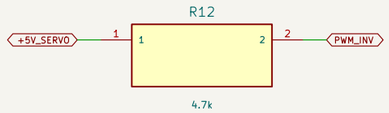


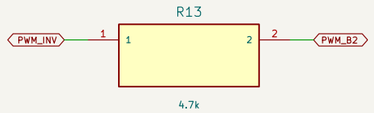


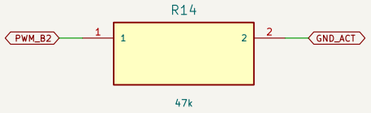


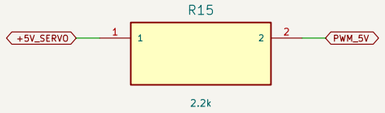


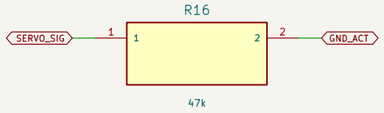


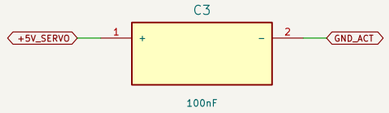


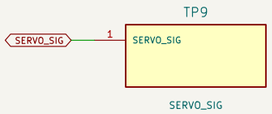


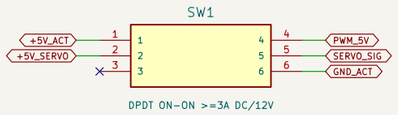














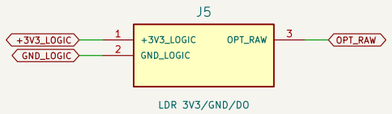
SW1 ON:1–2/4–5; OFF:2–3/5–6. Pin3 libre. Dos inversiones, PWM no invertido. J13 cable soldado.

Etiquetas globales del mismo nombre son la misma red. No autoriza soldadura.

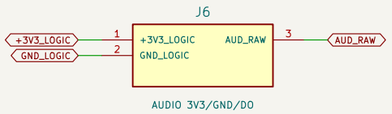
Sheet: /Servo y corte de mantenimiento/
File: 04_SERVO.kicad_sch

Title: Servo y corte de mantenimiento – Nodo A/B

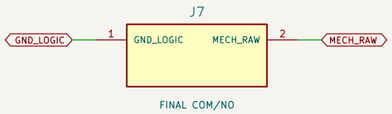
Size: A3	Date:	**Rev: T51-P1 PROVISIONAL**
KiCad E.D.A. 10.0.8		Id: 5/8



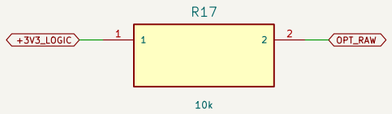
LDR 3V3/GND/DO



AUDIO 3V3/GND/DO



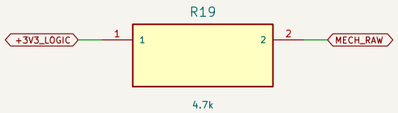
FINAL COM/NO



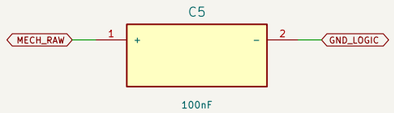
10k



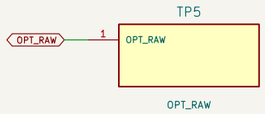
10k



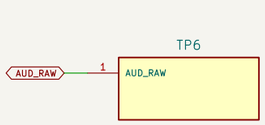
4.7k



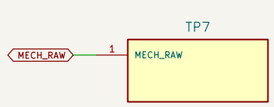
100nF



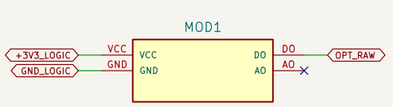
OPT_RAW



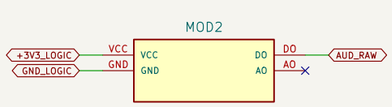
AUD_RAW



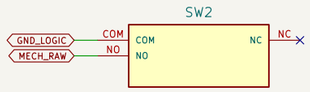
MECH_RAW



LDR LM393



KY037/HW484



SPDT rueda

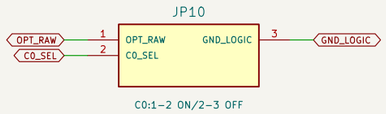
VCC de modulos SOLO 3.3V. Orden de pines se verifica por modulo. AO/NC libres. JP4–6 abiertos antes de medir.

Etiquetas globales del mismo nombre son la misma red. No autoriza soldadura.

Sheet: /Receptores y puntos de prueba/
File: 05_RX.kicad_sch

Title: Receptores y puntos de prueba — Nodo A/B

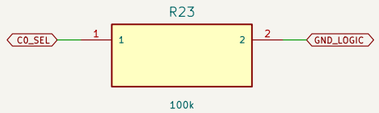
Size: A3	Date:	**Rev: T51–P1 PROVISIONAL**
KiCad E.D.A. 10.0.8		Id: 6/8



C0:1–2 ON/2–3 OFF



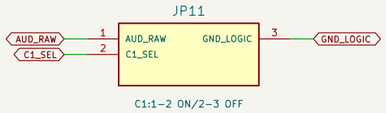
1k



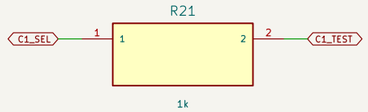
100k



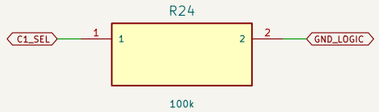
C0 LINK GPIO



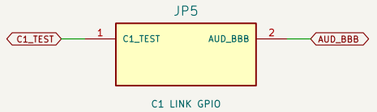
C1:1–2 ON/2–3 OFF



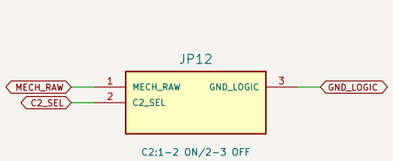
1k



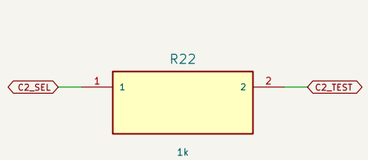
100k



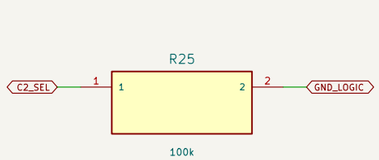
C1 LINK GPIO



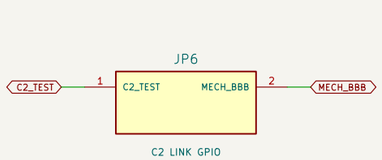
C2:1–2 ON/2–3 OFF



1k



100k



C2 LINK GPIO

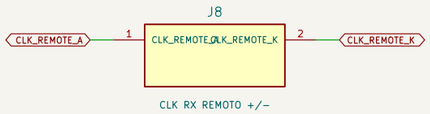
JP10/11/12 = EN_C0/1/2. 1–2 RX habilitado; 2–3 RX a cero. JP4–6: enlaces BBB, inicialmente abiertos.

Etiquetas globales del mismo nombre son la misma red. No autoriza soldadura.

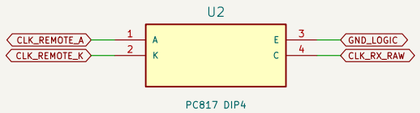
Sheet: /Selectores de fallo y enlaces GPIO/
File: 06_ENABLE.kicad_sch

Title: Selectores de fallo y enlaces GPIO — Nodo A/B

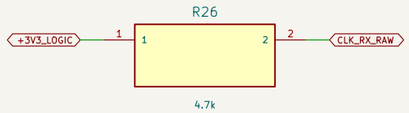
Size: A3	Date:	**Rev: T51–P1 PROVISIONAL**
KiCad E.D.A. 10.0.8		Id: 7/8



CLK RX REMOTO +/−



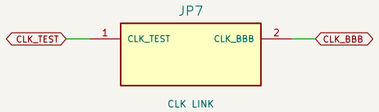
PC817 DIP4



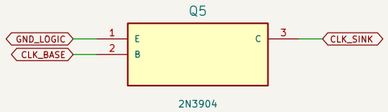
4.7k



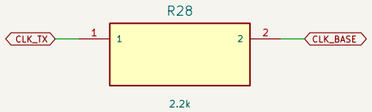
1k



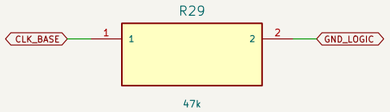
CLK LINK



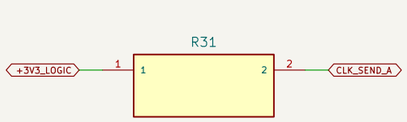
2N3904



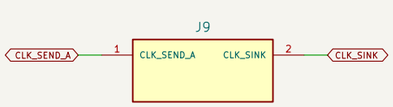
2.2k



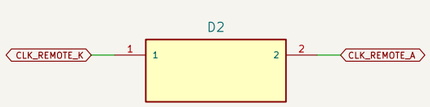
47k



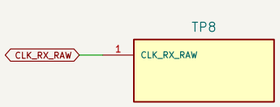
270



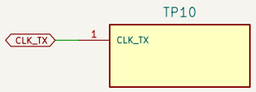
CLK TX +/−



1N4148



CLK_RX_RAW



CLK_TX

J9 A−>J8 B y J9 B−>J8 A por pares +/−; NO GND A−B. U2 RX activo bajo. JP7 abierto antes de medir.

Etiquetas globales del mismo nombre son la misma red. No autoriza soldadura.

Sheet: /Reloj aislado TX y RX/
File: 07_CLOCK.kicad_sch

Title: Reloj aislado TX y RX — Nodo A/B

Size: A3	Date:	**Rev: T51−P1 PROVISIONAL**
KiCad E.D.A. 10.0.8		Id: 8/8